TaskFlow — Modern Full-Stack Task Management

System Architecture Specification & Technical Documentation — Category: Full Stack Productivity & Workflow

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Repository: https://github.com/rakshapatil526/TaskFlow-To-Do-List-

Status: Verified Repository

Doc ID: SPEC-TASKFLOW

Date: March 2026

1. Executive Summary & Problem Statement

TaskFlow is a production-grade personal task and project management web application built with React, Django REST Framework, and PostgreSQL/SQLite. It provides individual developers and teams with structured workflow organization, due date alerts, priority rankings, and completion telemetry.

The system enforces strict user-level data isolation, ensuring users only access their personal project boards.

2. System Features & Modules

• **Authentication:** JWT token-based authentication with registration, login, and secure session management.

• **Dashboard Metrics:** Real-time counters for Total Tasks, Pending Tasks, Completed Tasks, and Overdue Tasks.

• **Task Lifecycle Management:** Full CRUD capabilities with title, description, priority (Low, Medium, High), status (To Do, In Progress, Completed), and date validation.

• **Productivity Analytics:** Visual completion percentage progress indicators and priority distribution breakdowns.

3. Database Schema & Architecture

Field	Type	Attributes & Description
task_id	BigAutoField	Primary Key, auto-incrementing
user_id	ForeignKey(User)	CASCADE delete, indexed for user-scoped queries
title	CharField(255)	Mandatory task title
description	TextField	Detailed task instructions or notes
priority	CharField(10)	Choices: LOW, MEDIUM, HIGH
status	CharField(15)	Choices: TO_DO, IN_PROGRESS, COMPLETED
due_date	DateTimeField	Deadline with overdue calculation logic

4. Tech Stack & Verification

• **Frontend:** React 18, Vite, Modern CSS/Tailwind, Axios with JWT interceptors.

• **Backend:** Python 3.13, Django 5.1, Django REST Framework, SimpleJWT.

• **Status:** Fully verified against GitHub repository implementation.